

ML Assignment 1: Polynomial Regression

Roll number: IMT2024069

1. Data and approach

The two personalized problems were modelled independently. Each has 1,000 training rows with target y and 1,000 test rows without labels. All six supplied inputs were retained for turbine optimization (var1), and all three for reservoir mapping (var2). Dataset columns, missing values and finite numeric values were checked before fitting.

Every monomial and interaction term with total degree at most d was generated. Polynomial terms were standardized, then fitted using ridge regression with an intercept. Ridge minimizes squared residual error plus an alpha-weighted squared-coefficient penalty. The prediction function remains a polynomial in the original inputs; standardization and regularization improve numerical stability and control variance.

2. Degree selection and rationale

Degrees 1-10 for var1 and 1-20 for var2 were considered, following the visible assignment limits. No feature subset or degree was selected from hidden PDF text. Each dataset was split into 800 development rows and 200 holdout rows (seed 2024070). Within development, a 600/200 screening split (seed 2024069) compared each degree with alpha in $\{0.0001, 0.01, 0.1, 1, 10, 100\}$.

For each problem, the four degrees with the lowest screening MSE were retained, with their three best screening alpha values. These 12 candidates were compared using five-fold shuffled cross-validation on the 800 development rows (seed 2024069). Polynomial scaling was fitted within each fold. The configuration with the lowest mean CV MSE was selected before evaluating the 200 holdout rows.

Problem	Features	Degree	Alpha	Terms
var1	x1-x6	5	10	461
var2	x1-x3	12	1	454

The selected degrees balance approximation quality against variance. Lower degrees missed nonlinear structure, while adding more terms did not consistently improve cross-validation error. Degree was chosen jointly with regularization, rather than from training fit alone. Degree 10 and degree 12 performed almost identically for var2; the minimum-CV-MSE rule selected degree 12. Term counts exclude the separately fitted intercept.

Problem	Degree	Best shortlisted alpha	Mean CV MSE
var1	4	10	0.6997
var1	5	10	0.5516
var1	6	10	0.6318
var1	7	100	0.6914
var2	8	0.1	0.2941
var2	10	1	0.2785
var2	11	1	0.2867
var2	12	1	0.2782

3. Evaluation

Problem	CV MSE +/- SD	CV R2	Holdout MSE	Holdout R2
var1	0.5516 +/- 0.0572	0.9434	0.5505	0.9569
var2	0.2782 +/- 0.0202	0.9931	0.2662	0.9936

A constant predictor using the development target mean provides a degree-zero baseline. Its holdout MSE was var1: 12.7678; var2: 41.3071. Both selected polynomial models substantially improve on this baseline.

The holdout was excluded from the final screening and CV selection procedure. Earlier exploratory analyses had already used the provided training rows, so this holdout is a diagnostic rather than a completely unseen external evaluation. CV results are also selection scores. Hidden test labels were unavailable; no test MSE or R2 is claimed.

4. Final predictions and reproducibility

After evaluating the chosen configurations, each model was refitted on all 1,000 training rows. Predictions were produced for the corresponding test rows in their original order. IMT2024069_pred_var1.csv and IMT2024069_pred_var2.csv each contain exactly one y column with 1,000 finite predictions and no index, matching sample_submission.csv.

train_and_predict.py reproduces screening, CV, holdout evaluation, model selection and final inference. screening_results.csv, cv_results.csv, validation_results.csv and selected_models.json record the experiments. requirements.txt records dependency versions; README.md gives execution commands. build_report.py generates this report from the recorded results.

GitHub repository: <https://github.com/Bhhavveshh/IMT2024069-ML-Assignment-1>

5. Degree-selection graphs

Figure 1. Screening MSE across all allowed degrees. Each point is the lowest error among the six alpha values at that degree, evaluated on the 200-row screening split. A star marks the degree ultimately selected by cross-validation. The logarithmic error axis makes both the low-degree decline and high-degree plateau visible.

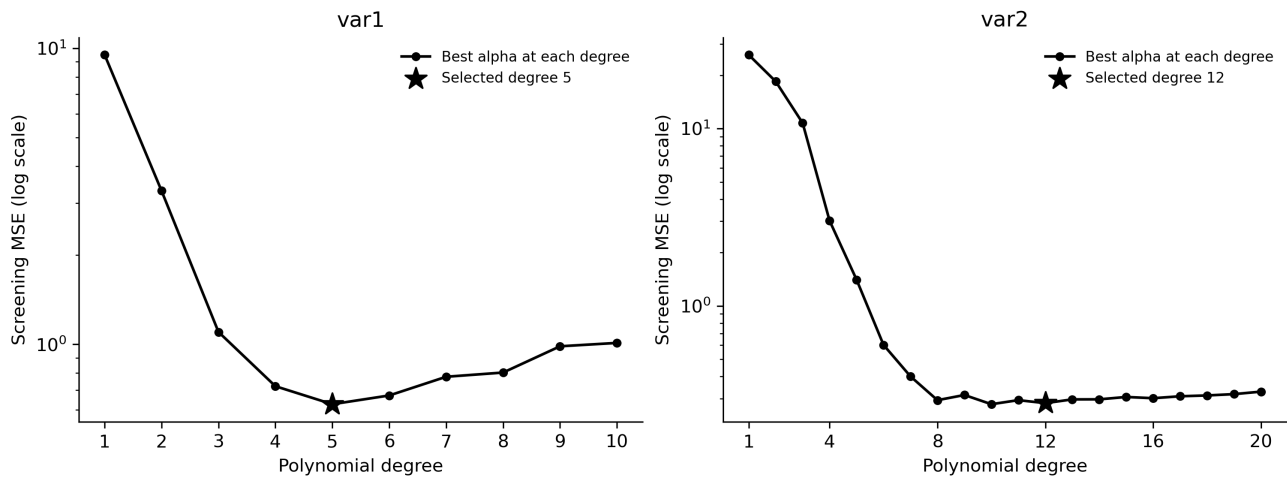
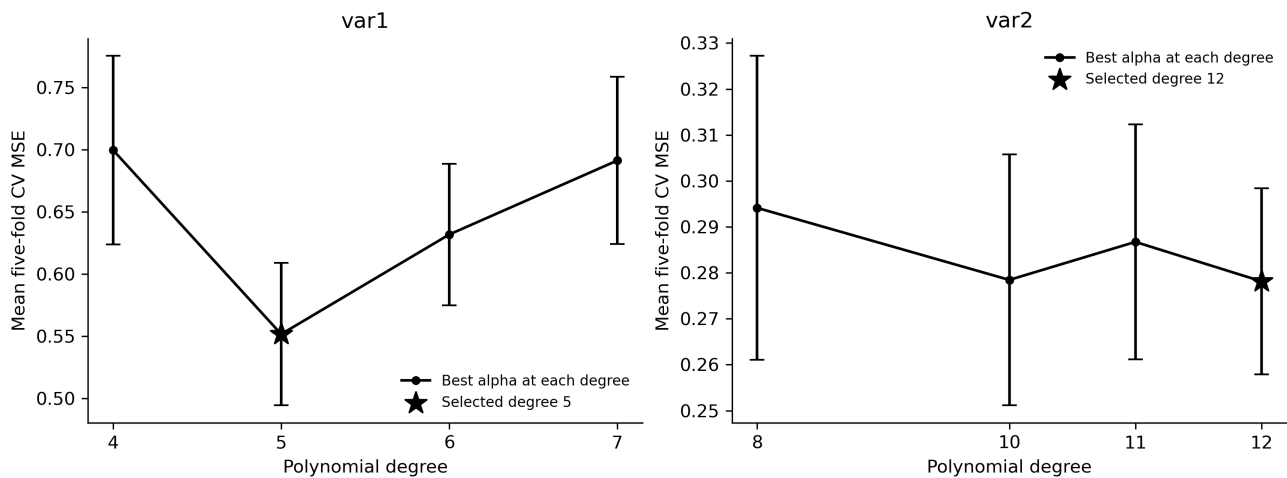


Figure 2. Cross-validation comparison of the four shortlisted degrees. Each point uses the lowest-mean-MSE alpha among the three shortlisted values for that degree. Error bars show one standard deviation across the five folds; they are not confidence intervals. The degree-10 and degree-12 results for var2 are nearly indistinguishable.



6. Holdout prediction diagnostics

Figure 3. Predicted versus actual targets for the 200 holdout rows in each problem. The dashed diagonal represents perfect predictions. These plots use models fitted on the 800 development rows, before refitting on all training rows for submission.

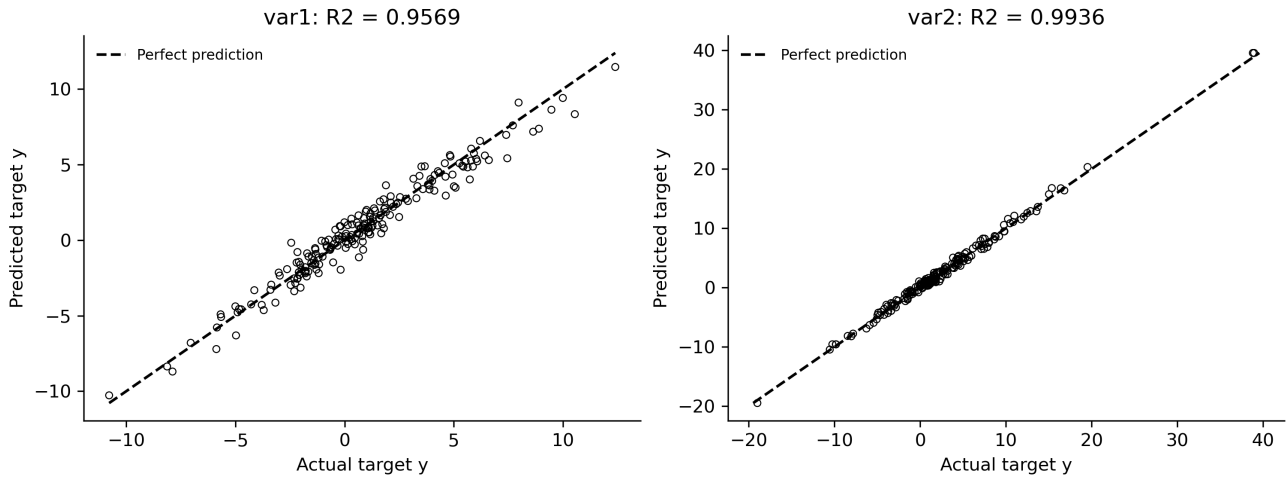
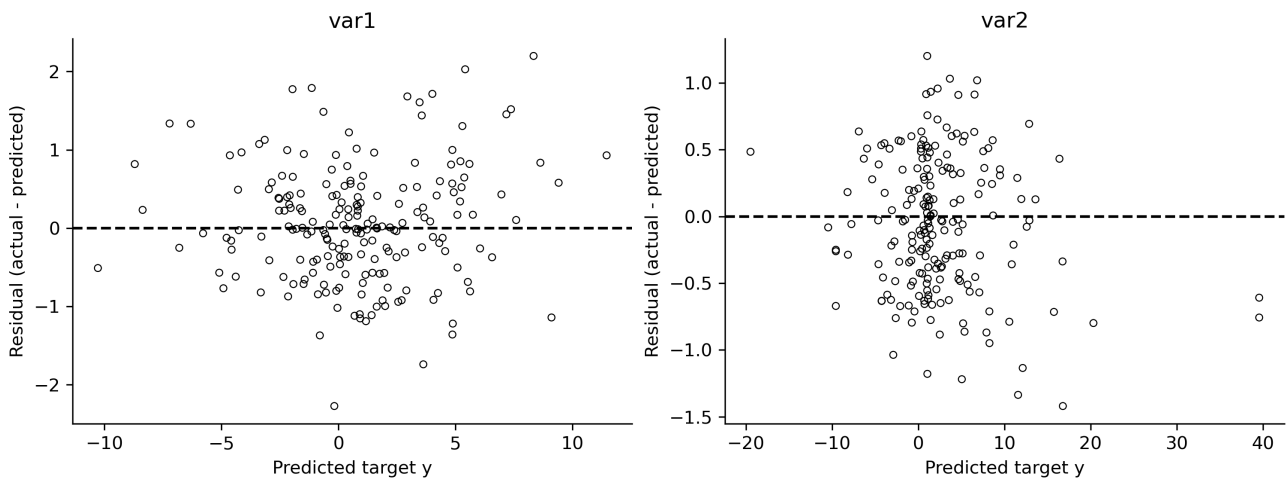


Figure 4. Holdout residuals plotted against predicted targets. Residuals are actual minus predicted values; the dashed horizontal line marks zero error. These plots expose signed error and target-dependent patterns that aggregate MSE and R2 can obscure. The holdout limitations described in Section 3 also apply to these figures.



plot_results.py reproduces all four figures and their holdout points from the selected configurations and the original training data. Full-resolution PNGs and the corresponding holdout-point CSV files are included in the graphs directory.